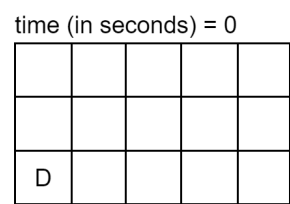
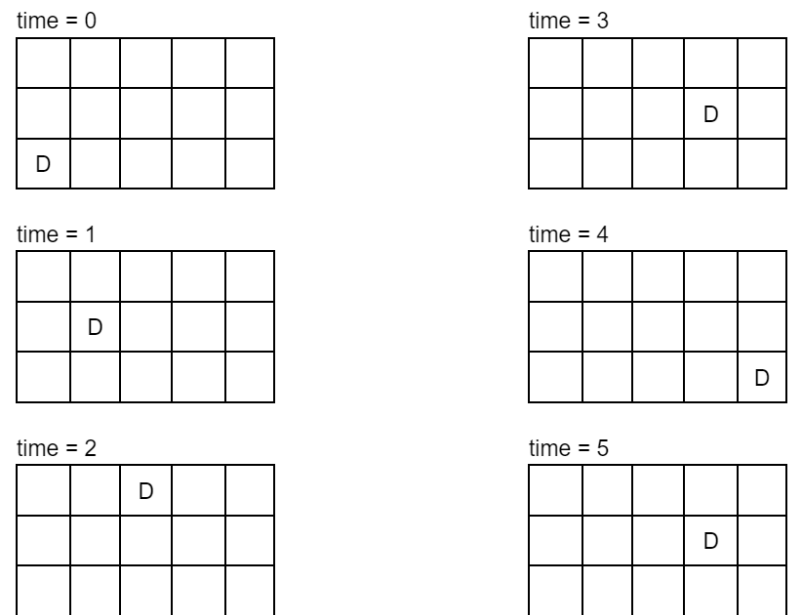


Evirir the dragon has a TV screen which can be represented by a 2D grid with h rows and w columns. The square on the r -th row from the bottom and c -th column from the left is denoted by (r, c) . A DVD screensaver logo, represented by a 1×1 square, starts at $(1, 1)$ and initially moves one square up and one square right per second.

Here is an example with $h = 3, w = 5$, where D is the logo:



When the logo hits exactly one border of the grid, it is reflected such that the angle of incidence is equal to the angle of reflection and continues with the same speed. When the logo hits a corner, it is reflected such that it moves in the opposite direction (and continues with the same speed).



After t seconds, on which square will the logo be on? If the square is (r, c) , output r c on a new line.

You are given Q independent test cases described by three integer arrays of length Q : H, W and T , where H_i, W_i and T_i are the values of h, w and t respectively for the i -th test case.

Note: For an array A , A_i denotes the i -th element of A .

Input Format

The input consists of an integer Q and three arrays of integers of length Q : H , W and T .

Constraints:

- $1 \leq Q \leq 1000$.
- $2 \leq H_i, W_i \leq 10^{12}$ ($1 \leq i \leq Q$)
- $0 \leq T_i \leq 10^{16}$ ($1 \leq i \leq Q$)

Output Format

For each test case, in the original order, if the answer is (r, c) , output `r c` (separated by one space) on a new line.

Sample Input

```
Q = 4
H = [3, 2, 7, 36]
W = [5, 2, 2, 28]
T = [5, 5, 0, 127]
```

Sample Output

```
2 4
2 2
1 1
14 20
```

Explanation

The first test case is described in the statement.

For the second test case, the logo visits the squares $(1, 1), (2, 2), (1, 1), (2, 2), (1, 1), (2, 2)$ in that order.

For the third test case, $t = 0$, so the square will be on its starting square, $(1, 1)$.

Evirir sang naga mempunyai suatu skrin TV yang boleh diwakili oleh suatu grid 2D dengan h baris dan w lajur. Petak pada baris ke- r dari bawah dan lajur ke- c dari kiri ditulis sebagai (r, c) . Suatu logo `screensaver` DVD, yang diwakili oleh suatu petak 1×1 , bermula pada $(1, 1)$ dan mula bergerak satu petak ke atas dan satu petak ke kanan pada setiap saat.

Berikut merupakan suatu contoh dengan $h = 3$, $w = 5$, di mana `D` ialah logo tersebut:

time (in seconds) = 0

D				

Apabila logo tersebut bersentuh dengan satu sisi grid tersebut, ia dipantulkan supaya sudut tujuhnya adalah sama dengan sudut pantulan, dan ia terus bergerak dengan kelajuan yang sama. Apabila logo tersebut bersentuh dengan suatu bucu, ia dipantulkan supaya ia bergerak pada arah bertentangan (dan terus bergerak dengan kelajuan yang sama).

time = 0

D				

time = 3

			D	

time = 1

	D			

time = 4

				D

time = 2

		D		

time = 5

			D	

Selepas t saat, petak manakah yang akan diduduki oleh logo tersebut? Jika petak tersebut ialah (r, c) , output `r c` pada baris yang baru.

Anda diberi Q kes ujian yang bebas, diuraikan dengan tiga tatasusunan (array) integer dengan panjang Q : H , W dan T , di mana H_i , W_i dan T_i masing-masing adalah nilai bagi h , w dan t

bagi kes ujian ke- i .

Nota: Bagi suatu tatasusunan A , A_i melambangkan unsur ke- i bagi A .

Format Input

Input terdiri daripada satu integer Q dan tiga tatasusunan integer dengan panjang Q : H , W dan T .

Kekangan:

- $1 \leq Q \leq 1000$.
- $2 \leq H_i, W_i \leq 10^{12}$ ($1 \leq i \leq Q$)
- $0 \leq T_i \leq 10^{16}$ ($1 \leq i \leq Q$)

Format Output

Bagi setiap kes ujian, dalam susunan asal, jika jawapannya ialah (r, c) , output `r c` (diasingkan dengan satu ruang) pada baris yang baru.

Contoh Input

```
Q = 4
H = [3, 2, 7, 36]
W = [5, 2, 2, 28]
T = [5, 5, 0, 127]
```

Contoh Output

```
2 4
2 2
1 1
14 20
```

Penjelasan

Kes ujian pertama telah dihuraikan dalam pernyataan soalan ini.

Bagi kes ujian kedua, logo tersebut melalui petak-petak

$(1, 1), (2, 2), (1, 1), (2, 2), (1, 1), (2, 2)$ mengikut urutan tersebut.

Bagi kes ujian ketiga, $t = 0$, maka logo tersebut akan berada pada petak asalnya, $(1, 1)$.

Solution

The horizontal and vertical components are independent from each other, so we can simply solve them separately. Now we only need to solve the 1D version.

For the horizontal component, note that the column of the logo's location repeats every $2(W - 1)$ seconds. Therefore, we can use the value of T in modulo $2(W - 1)$ instead (i.e. the remainder when T is divided by $2(W - 1)$).

Let $X = T \% (2(W - 1))$. The distance of the logo from the W -th column at time t is $|(W - 1) - X|$ where $0 \leq t \leq 2(W - 1) - 1$. Hence, the answer for the horizontal component is $W - |(W - 1) - X|$.

The answer for the vertical component is similar.